ChE 2O04 Tutorial 3

Tutorial Problem #1

A 10 cm diameter steel ball is suspended by a string in a vessel filled with two liquids, water and carbon tetrachloride. The fluids are immiscible and form two layers: the bottom layer is 30 cm deep, and the top layer is 35 cm deep.

a) Draw a diagram of the process with the two layers labelled as the appropriate fluids.

b) Draw a force diagram on the ball (hint: there are three forces due to gravity, buoyancy, and tension)

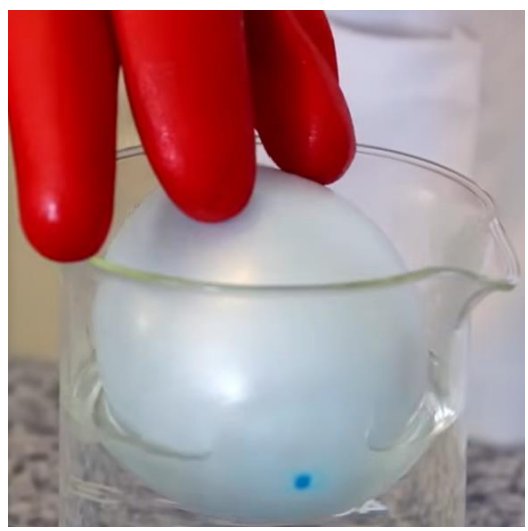
c) Calculate the tension in the string when the ball is completely submerged in carbon tetrachloride.

d) Calculate the tension in the string when the ball is submerged half in water and half in carbon tetrachloride.

e) Calculate the tension in the string when the ball is submerged half in the top layer and half in air.

Tutorial Problem #2

The pressure of a gas is a function of its density and temperature, as described by the ideal gas law (Eq. 1.1 in the textbook). A good example of this is a balloon exposed to hot and cold temperatures, as shown in the attached video: (https://www.youtube.com/watch?v=NplVuTrr59U&t=158s)



In lecture it was shown that for the special case of isothermal (constant temperature), the resulting equation for pressure variation was:

$$P_2 = P_1 \exp\left[-\frac{gM_w}{R_U T}(z_2 - z_1)\right]$$

a) Following a similar approach to that presented in lecture for the isothermal case, derive the general expression for pressure as a function of elevation for the general case of linear temperature variation:

$$T = mz + T_o$$

b) The special case of linear temperature variation with height is exactly what happens in the troposphere, the lowest level of Earth's atmosphere. Using the data in the table below, calculate the value of the constant m in the equation:

$$T = mz + T_o$$

Elevation (metres)	Temperature (Kelvin)
0	288.15
11,000	216.65

Hints: The constant m in the equation above is the best-fit value of the slope of temperature versus elevation.
T_o is the temperature at sea level (i.e. z = 0)

c) Now substitute the value for m found in Part B into the formula derived in Part A and use the resulting expression to make a plot of atmospheric pressure versus elevation.

- Set the x-axis range from 0 to 9,000 m
- Set the y-axis range from 0 to 120 kPa

Hint: Use the following values in your calculation:

M_w = 28.97 kg/kg-mol
R_u = 8314 J/kg-mol/K

d) Justify and check the plot in part C through the following statement: "*the atmospheric pressure at the summit of Mount Everest (~8,800 m) is roughly 1/3 of that at sea level".*